WORKSHOP · DAY 02 · CHEAT SHEET

# From a model to a measurement. The five-step proxy workflow.

WORKSHOP MATERIALS
GITHUB.COM/JACQPARK/
**INSTATS-LLM-WORKSHOP**

## THE PIPELINE · FROM TEXT TO REGRESSION-READY CSV

01
### Codebook
3 rules, operational, disjoint, testable. Spend longer here than on the coding.

→

02
### Gold set
30 paragraphs, stratified. Two coders if you can. Disagreements re-read.

→

03
### Hypothesis ensemble
3+ hypotheses, weighted. Score = weighted average of entailment probs.

→

04
### Validate
Weighted F1, Pearson r, confusion matrix. Re-read off-diagonals.

→

05
### Aggregate
Group by country-year. Pick mean, length-weighted, or top-K. Plot.

## CODEBOOK · THREE RULES THAT SEPARATE CODEBOOK FROM WISH

RULE 01
### Operational
Each rule names a feature you can find in the text. "Mentions Watch List" works. "Has a critical tone" does not.

RULE 02
### Disjoint
No paragraph fits two rules. Praise + one concern resolves to exactly one bin.

RULE 03
### Testable
Two coders applying the rules to the same paragraph reach the same answer most of the time.

## VALIDATION METRICS · FOLLOWING PARK (2026)

| METRIC | WHAT IT MEASURES | MEASURED OUTPUTS |
|---|---|---|
| **Weighted F1** | Class-balanced F1 | 0.787 |
| **Accuracy** | Diagonal of confusion matrix | 0.773 |
| **Pearson r** | Continuous score vs gold | 0.676 |
| **Spearman ρ** | Rank correlation | 0.689 |

### Aggregation choices

`Mean`
Simple. Treats every paragraph as equally informative.

`Length-weighted`
Verbose paragraphs count more. Re-read your data first.

`Top-K`
Keep the K most extreme paragraphs. Validate K against gold.

## HYPOTHESIS ENGINEERING · THE RECIPE

**Start simple.** Three hypotheses. One positive, one negative, one for the residual class. Use `multi_label=True` so each hypothesis returns its own entailment probability. **Score formula.** `score = sum(weight_i × prob_i) / sum(|weight_i|)`. **Add hypotheses in response to gold-set misses.** Re-read 10 disagreements. Group by failure mode. For each mode, add one targeted hypothesis. Set weights by codebook judgment, not by tuning. **Stop adding hypotheses** when weighted F1 stops improving. More than 15 rarely helps.

## CODE · TWO SNIPPETS TO KEEP

ENSEMBLE · WEIGHTED ENTAILMENT SCORES

```
from transformers import pipeline
import numpy as np

zsc = pipeline("zero-shot-classification",
  model="MoritzLaurer/DeBERTa-v3-"
        "base-mnli-fever-anli")
hyps = ["text praises target",
        "text criticizes target"]
w = np.array([+1.0, -1.0])

p = zsc(text, hyps, multi_label=True)
prob = np.array([
  p["scores"][p["labels"].index(h)]
  for h in hyps])
score = (prob*w).sum() / np.abs(w).sum()
```

VALIDATION · F1, CORRELATION, CONFUSION

```
from sklearn.metrics import (
  f1_score, accuracy_score, confusion_matrix)
from scipy.stats import pearsonr, spearmanr

f1  = f1_score(gold, pred, average="weighted")
acc = accuracy_score(gold, pred)
r,_ = pearsonr(score, gold)
ρ,_ = spearmanr(score, gold)
cm  = confusion_matrix(gold, pred)

print(f"F1 {f1:.3f}  r {r:.3f}")
```

## VALIDATION APPENDIX · WHAT TO SHIP

**Codebook**
The full text of every rule. The dated version that produced the gold set.

**Gold set**
Sample size, sampling method, number of coders, agreement statistic if two coders.

**Model and hypotheses**
HF model ID with version pin, every hypothesis, every weight.

**Validation numbers**
Weighted F1, accuracy, Pearson r, Spearman ρ against gold. Confusion matrix.

**Replication package**
Notebook, gold-set CSV, scored-paragraph CSV, aggregated CSV.

## PITFALLS

- **Skipping the gold set.** No gold means no F1 means no defense. Code 30 paragraphs.
- **Tuning weights to the gold.** Weights come from the codebook. If you train them, you need a held-out set.
- **Single hypothesis on mixed text.** The model picks the dominant signal. Use an ensemble.
- **Accuracy alone.** A class-imbalanced corpus produces 80% accuracy from a constant predictor. Always report weighted F1.
- **Mean aggregation on uneven coverage.** A country-year with one paragraph is not the same as one with forty. Always report n.
- **Forgetting subgroups.** Headline F1 hides drift. Stratify by region or time period in the appendix.